

Measures of Skewness and Kurtosis, like measures of central tendency and dispersion, study the characteristics of a frequency distribution. Averages tell us about the central value of the distribution and measures of dispersion tell us about the concentration of the items around a central value.

What is Symmetry (normal distribution)?

In mathematics, the quality of having two parts that match exactly, either when one half is like an image of the other half in a mirror.

Why is the normal distribution important?

Normal distribution is so important is that it is easy for mathematical statisticians to work with. This means that many kinds of statistical tests can be derived for normal distributions. Fortunately, these tests work very well even if the distribution is only approximately normally distributed. Some tests work well even with very wide deviations from normality.

What is Skewness?

Skewness means the study of a curve. Skewness can be quantified to define the extent to **which a distribution differs from a normal distribution.**

We can observe the Skewness of the curve whether it is **positively Skewed** or **negatively Skewed** or **zero Skewed**

So, skewness indicates whether the observations in a data set are **concentrated on one side.**

What is 'Kurtosis'?

When two or more symmetrical distributions (normal distribution) are **compared**, the difference in them is studied with 'Kurtosis'.

Kurtosis is a measure of whether the data are heavy-tailed or light-tailed relative to a normal distribution. That is, data sets with high kurtosis tend to have heavy tails, or outliers. Data sets with low kurtosis tend to have light tails, or lack of outliers.

Dealing with Skewness and Kurtosis:

Many classical statistical tests and intervals depend on normality assumptions.

Significant skewness and kurtosis clearly indicate that data are not normal. If a data set exhibits significant skewness or kurtosis (as indicated by a histogram or the numerical measures), what can we do about it?

One approach is to apply some type of **transformation to try to make the data normal, or more nearly normal.** The [Box-Cox transformation](#) is a useful technique for trying to normalize a data set. In particular, taking the log or square root of a data set is often useful for data that exhibit moderate right skewness.

Tests of Skewness:

There are certain tests to know whether skewness does or does not exist in a frequency distribution.

They are:

1. In a skewed distribution, values of mean, median and mode would not coincide. The values of mean and mode are pulled away and the value of median will be at the centre.
2. Quartiles will not be equidistant from median.(IQR is not equal)
3. When the asymmetrical distribution is drawn on the graph paper, it will not give a bell shaped curve.
4. Sum of the positive deviations from the median is not equal to sum of negative deviations.(the data points in the left side median are not equal to the right side of the median).

Nature of Skewness:

Skewness can be positive or negative or zero.

1. When the values of mean, median and mode are equal, there is **no**

Skewness(normal).

2. When $\text{mean} > \text{median} > \text{mode}$, skewness will be positive.
3. When $\text{mean} < \text{median} < \text{mode}$, skewness will be negative.

Methods of ascertaining Skewness:

Skewness can be studied graphically and mathematically. When we study skewness graphically, we can find out whether skewness is positive or negative or zero. This can be shown with the help of a diagram:

Mathematically skewness can be studied as:

(a)Absolute-Skewness

$$\text{Absolute skewness} = Q_3 + Q_1 - 2 \text{ Median}$$

When the skewness is presented in absolute term i.e, in units, it is absolute skewness.

(b)Relative or coefficient of skewness

If the value of skewness is obtained in ratios or percentages, it is called relative or coefficient of skewness. Relative measures of skewness is also called coefficient of skewness.

Mathematical measures of skewness can be calculated by :

1. **Bowley's Method:**
2. **Karl-Pearson's Method(Coefficient of skweness):**
3. **Kell's Measure of Skewness:**

Bowley's method of skewness is based on the values of median, lower and upper quartiles.

Wherever positional measures are given, skewness should be measured by Bowley's method. This method is also used in case of 'open-end series', where the importance of extreme values is ignored. (because we can find Bowley's Method by using quartiles. Quartiles is not affected with large value similarly bowleys is also not affected with large values). **Coefficient of skewness lies within the limit ± 1 .**

Bowley's Skewness $= (Q1 + Q3 - 2Q2) / (Q3 - Q1)$.

# of pets	# of families	cumulative freq
0	60	60
1	60	120
2	50	170
3	20	190
4	25	215
5	10	225
6 or more	5	230

Step 1: Find the [Quartiles](#) for the data set. You'll want to look for the "nth" observation using the following formulas:

$$Q1 = (\text{total cum freq} + 1 / 4)\text{th observation} = (230 + 1 / 4) = 57.75$$

$$Q2 = (\text{total cum freq} + 1 / 2)\text{th observation} = (230 + 1 / 2) = 115.5$$

$$Q3 = 3 (\text{total cum freq} + 1 / 4)\text{th observation} = 3(230 + 1 / 4) = 173.25$$

Step 2: Look in your table to find the nth observations you calculated in Step 1:

$$Q1 = 57.75\text{th observation} = 0$$

$$Q2 = 115.5\text{th observation} = 1$$

$$Q3 = 173.25\text{th observation} = 3$$

Step 3: Plug the above values into the formula:

$$Skq = Q3 + Q1 - 2Q2 / Q3 - Q1$$

$$Skq = 3 + 0 - 2 / 3 - 0 = 1/3.$$

$Skq = + 0.3333$, so the distribution is positively skewed. or in another words the departs value of the distribution to a normal distribution is 0.3333.

Karl Pearson has suggested two formulae;

(i) where the relationship of mean and mode is established;

Pearson's Coefficient of Skewness #1 uses the [mode](#). The formula is:

$$SK = (\text{Mean} - \text{Mode}) / \sigma$$

Sample problem: You have data with a mean of 19, a mode of 20 and a standard deviation of 25. What does Pearson Mode Skewness tell you about the distribution?

$$Sk = (\text{mean} - \text{mode}) / \sigma = (19 - 20) / 25 = -0.04.$$

There is a very slight negative skewness (-0.04).

(ii) where the relationship between mean and median is not established.

Pearson's Coefficient of Skewness #2 uses the [median](#). The formula is:

$$Sk = \text{Mean} - (3\text{Median} - 2\text{Mean}) / \sigma$$

$$= \text{Mean} - 3\text{Median} + 2\text{Mean} / \sigma$$

$$= 3\text{Mean} - 3\text{Median} / \sigma$$

$$SK = 3(\text{Mean} - \text{Median}) / \sigma$$

If you don't know the mode, you won't be able to use Pearson mode skewness. However, the direction of skewness can be also figured out by finding where the [mean](#) and the [median](#) are.

Where we can use mode formula and where we can use median formula?

Caution: Pearson's first coefficient of skewness uses the mode. Therefore, if the mode is made up of too few pieces of data it won't be a stable measure of central tendency. For example, the mode in both these sets of data is 9:

1 2 3 4 5 6 7 8 9 9.

1 2 3 4 5 6 7 8 9 9 9 9 9 9 9 9 9 9 9 10 12 12 13.

In the first set of data, the mode only appears twice. This isn't a good measure of [central tendency](#) so you would be cautioned *not* to use Pearson's coefficient of skewness. The second set of data has a more stable set (the mode appears 12 times). Therefore, *Pearson's coefficient of skewness will likely give you a reasonable result.*

Pearson Skewness: What the Results mean?

The difference between the mean and mode, or mean and median, will tell you how far the distribution departs from symmetry. . A symmetric distribution (for example, the [normal distribution](#)) has a skewness of zero.

Kelly's Measure of Skewness is one of several ways to measure [skewness](#) in a data distribution. [Bowley's skewness](#) is based on the middle 50 percent of the observations in a data set. It leaves 25 percent of the observations in each tail of the distribution. Kelly suggested that leaving out fifty percent of data to calculate [skewness](#) was too extreme. **He created a measure to find skewness with more data.** Kelly's measure is based on P90 (the 90th [percentile](#)) and P10 (the 10th percentile). Only **twenty percent of observations** (ten percent in each tail) are excluded from the measure.

Kelly's Measure Formula:

Kelley's measure of skewness is given in terms of percentiles and [deciles](#)(D). Kelley's absolute measure of skewness (Sk) is:

$$Sk = P90 + P10 - 2 * P50 = D9 + D1 - 2 * D5.$$

This formula is not practically used. In fact, Kelly's measure of skewness is rarely used at all, even in its more common form, which is measured as coefficient of skewness:

What does Kelly's Measure of Skewness Tell us?

Kelly's Measure of Skewness gives you the same information about skewness as the other three types of skewness measures (Bowley's Skewness, Pearson Mode Skewness and Momental Skewness.).

“By using these three methods we get some answers. The answer will tell you how far the distribution departs from symmetry.”

The measure of central tendency, dispersion and skewness may not give a complete idea about the distribution. Another statistical measure is required to know about flatness, peakedness of the curve called kurtosis.

What is Kurtosis?

This heaviness or lightness in the tails usually means that your data looks flatter (or less flat) compared to the normal distribution. The standard normal distribution has a kurtosis of 3, so if your values are close to that then your graph is nearly normal. These nearly [normal distributions](#) are called **mesokurtic**.

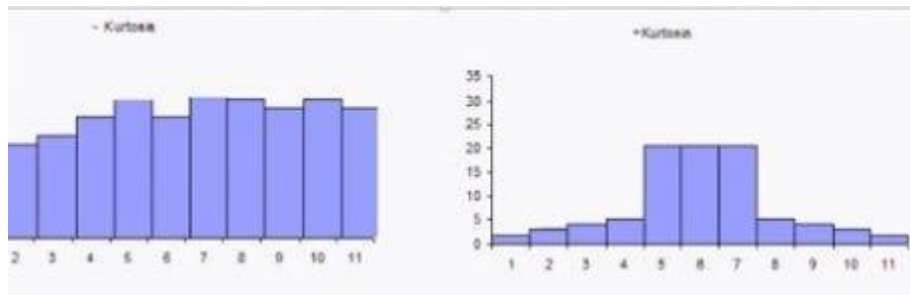
1. Mesokurtic:

Mesokurtic distributions are technically defined as having a kurtosis of zero, although the distribution doesn't have to be *exactly* zero in order for it to be classified as mesokurtic. The most common mesokurtic distributions are:

- The normal distribution.
- Any distribution with a Gaussian (normal) shape and zero
- The binomial distribution is mesokurtic for some values (i.e. for $p = \frac{1}{2}$)

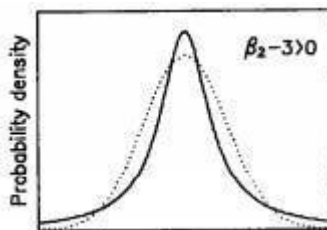
2. Platykurtic:

Platykurtic distributions have negative kurtosis. The tails are very thin compared to the normal distribution.



3. Leptokurtic:

A leptokurtic distribution has excess positive kurtosis, where the kurtosis is greater than 3. The tails are fatter than the normal distribution. The following illustration shows a leptokurtic distribution along with a normal distribution (dotted line).

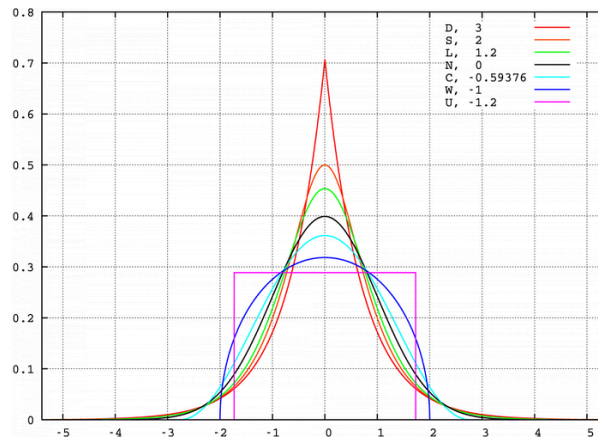


The Student's t-test is an example of a leptokurtic distribution. The t-distribution has fatter tails than the normal (you can also look at the first image above to see the fatter tails). Therefore, the critical values in a Student's t-test will be larger than the critical values from a z-test.

What is Excess kurtosis?

Excess kurtosis is *usually* defined as $kurt - 3$ (see Important note about equations). It is a measure of how the distribution's tails compare to the normal (Aldrich, E, 2014).

- Excess kurt for the normal distribution is 0 (i.e. $3 - 3 = 0$).
- Negative excess equals lighter tails than a normal distribution.
- Positive excess equals heavier tails than the normal.



Red, kurt 3, Laplace (D)ouble exponential distribution;
 Orange, kurt 2, hyperbolic (S)ecant distribution;
 Green, kurt 1.2, (L)ogistic distribution;
 Black, kurt 0, (N)ormal distribution;
 Cyan, kurt -0.593762..., raised Cosine distribution;
 Blue, kurt -1, (W)igner semicircle distribution;
 Magenta, kurt -1.2, (U)niform distribution.

What is the coefficient of skewness?

The coefficient of skewness is a measure of asymmetry in the distribution. A positive skew indicates a longer tail to the right, while a negative skew indicates a longer tail to the left. A perfectly symmetric distribution, like the normal distribution, has a skew equal to zero.

● Karl Pearson's Coefficient: -

Karl Pearson's Coefficient is a mathematical tool that helps in signifying a relationship within two different variables that are calculated under the same scale or ratio.

Karl Pearson's Coefficient of Correlation is a mathematical approach in which the numerical expression is used to estimate or determine the range or magnitude and the direction of the relationship between two linearly related variables. Karl Pearson's Coefficient is a quantitative method that is often used in statistics. It is also known as the Pearsonian Coefficient of Correlation, and it is a widely accepted and widely used method. To calculate the measurement of the relationship between two variables Karl Pearson's Coefficient of Correlation formula is used.

How to calculate Karl Pearson's Coefficient of Correlation?

Karl Pearson's Coefficient of Correlation formula is used to obtain accurate results. The formula includes denoting the coefficient of correlation as 'r'. The complete formula is:

$$R = (n(\sum xy) - (\sum x)(\sum y)) / (\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]})$$

Where

R = Pearson Coefficient

n = number of the pairs of the stock

$\sum xy$ = sum of products of the paired stocks

$\sum x$ = sum of the x scores

$\sum y$ = sum of the y scores

$\sum x^2$ = sum of the squared x scores

$\sum y^2$ = sum of the squared y scores

Karl Pearson's Coefficient of skewness

Karl Pearson's Coefficient of Skewness method is another method that was made by Karl Pearson. This method is performed by using mean and mode, which tells that Karl Pearson's Coefficient of skewness is calculated based on descriptive statistical methods.

Karl Pearson's Coefficient of Skewness formula

There are two different ways of calculating skewness through using Karl Pearson's Skewness formula.

1- Karl Pearson's Skewness formula through the usage of mode

– (Mean–Mode)/ S.D

2- Karl Pearson's Skewness formula through the usage of the median

– 3(Mean–Median)/ S.D

Use of Karl Pearson's Coefficient Correlation

Karl Pearson's Coefficient of Correlation has many uses in the practical field. The following is a list that states the application of Karl Pearson's Coefficient of Correlation in the practical field.

Business Problems- The Karl Pearson coefficient of correlation method proves to be useful for banks. It helps a bank to identify or determine the correlation between the credit card delinquency rate of a card beholder and their income.

Input Data- Karl Pearson's coefficient of correlation method is also used in the input of data. It can be used to input the data of the monthly income of each credit card customer and the delinquency rate of each credit card customer.

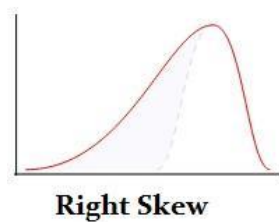
Business Benefit- Karl Pearson's coefficient of correlation formula gives various benefits to a business. By using the above example, we can say that Karl Pearson's

coefficient of correlation formula also helps the credit card manager to decide an individual limit of credit by calculating the correlation coefficient between delinquency rates and income.

- **Bowley's coefficient of skewness**

It is also known as quartile skewness coefficient or Bowley's skewness coefficient. It is calculated using the formula $(Q3 + Q1 - 2Q2) / (Q3 - Q1)$, where $Q1$, $Q2$, and $Q3$ are the first, second, and third quartiles, respectively.

Bowley skewness is a way to figure out if you have a positively-skewed or negatively skewed distribution. One of the most popular ways to find skewness is the Pearson Mode Skewness formula. However, in order to use it you must know the mean, mode (or median) and standard deviation for your data. Sometimes you might not have that information; Instead you might have information about your quartiles. If that's the case, you can use Bowley Skewness as an alternative to find out more about the asymmetry of your distribution. It's very useful if you have extreme data values (outliers) or if you have an open-ended distribution.



$$\text{Bowley Skewness} = (Q3 + Q1 - 2Q2) / (Q3 - Q1)$$

Skewness = 0 means that the curve is symmetrical.

Skewness > 0 means the curve is positively skewed.

Skewness < 0 means the curve is negatively skewed.

Bowley Skewness Worked Example

Q. Find the Bowley Skewness for the following set of data:

bowley skewness example

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Step 2: Look in your table to find the nth observations you calculated in Step 1:

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Step 3: Plug the above values into the formula:

$$Sk_q = (Q_3 + Q_1 - 2Q_2) / (Q_3 - Q_1)$$

$$Sk_q = (3 + 0 - 2) / (3 - 0) = 1/3$$

$Sk_q = + 1/3$, so the distribution is positively skewed.

Why Bowley Skewness works.

In a symmetric distribution, like the normal distribution, the first (Q_1) and third (Q_3) quartiles are at equal distances from the mean (Q_2). In other words, $(Q_3 - Q_2)$ and $(Q_2 - Q_1)$ will be equal. If you have a skewed distribution then there will be a difference between those two values.

Limitations of Bowley Skewness.

Bowley Skewness is an absolute measure of skewness. In other words, it's going to give you a result in the units that your distribution is in. That's compared to the Pearson Mode Skewness, which gives you results in a dimensionless unit — the standard deviation. This means that you cannot compare the skewness of different distributions with different units using Bowley Skewness.

Calculate Bowley's coefficient of skewness for the following distribution of weekly wages of workers.

Wages	Below 300	300 — 400	400 — 500	500 — 600	600 — 700	Above 700
No. of Workers	5	8	18	35	27	7

Class	Frequency	Less Than Type cumulative frequency
Below 300	5	5
300–400	8	13
400–500	18	31
500–600	35	66
600–700	27	93
Above 700	7	100
Total	N=100	

Medium

Q_1 = value of $(4N)$ th observation:

= value of 25th observation.

25th observation belongs to the class 400–500.

Its frequency(f) is equal to 18 and cumulative frequency of the previous class is equal to 13.

$$Q1 = L + fh.(N/4 - C.F.)$$

$$= 400 + 18(100)(25 - 13)$$

$$= 466.67$$

Q2 = value of (2N)th observation

= value of 50th observation.

50th observation is from class 500–600 f=35 and C.F.=31

$$Q1 = L + fh.(N/2 - C.F.)$$

$$= 500 + 35(100)(50 - 31)$$

$$= 554.28$$

Q3 = value of (43N)th observations

= value of 75th observation

75th observation belongs to class 600 to 700

$$f=27, C.F.=66$$

$$Q3 = L + fh.(3N/4 - C.F.)$$

$$= 27(100)(75 - 66)$$

$$= 633.33$$

Bowley's coefficient of skewness is given by

$$S_{kb} = \frac{Q3 - Q1}{Q3 + Q1 - 2Q2}$$

$$= \frac{633.33 - 466.67}{633.33 + 466.67 - 2(554.28)}$$

$$= \frac{166.66}{1100 - 1108.56}$$

$$= -0.05$$